Jaehyun Ha

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Research Interests

Building AI-native database systems for unified analytics and inference on multi-modal data

- **Graph Databases:** Architecting a foundational query framework based on a unified graph representation of multi-modal data
- **Semantic Operator:** Embedding Large Language Models (LLMs) into the database to enable context-aware querying of unstructured data
- **Query Optimization:** Designing proxy models and novel cardinality estimation methods to optimize the execution of computationally expensive semantic operators

Education

POSTECH, Graduate School of Artificial Intelligence — Mar. 2022 – Present
M.S./Ph.D. Integrated Program — (Expected: Feb. 2028)
Data Systems Lab
Advisor: Prof. Wook-Shin Han

POSTECH, Dept. of Computer Science and Engineering — Mar. 2018 – Mar. 2022
B.S. in Computer Science and Engineering
GPA: 4.06 / 4.3
*Graduated with the **highest** GPA in CS Dept.*

Ulsan Science High School — Mar. 2016 – Mar. 2018

Professional Experiences

Google Systems Research Group, Sunnyvale, CA, USA — Jun 2026 – Sep 2026
Visiting Student Researcher

- Researching hardware-software co-optimization to accelerate semantic queries
- Hosted by Yeounoh Chung

University of Illinois Urbana-Champaign (UIUC), IL, USA — Jun 2025 – Sep 2025
Visiting Scholar

- Conducted research on semantic operator optimization, resulting in a paper accepted at SIGMOD 2027
- Joint research project under the supervision of Prof. Yongjoo Park

Digital Platform team, SK hynix Inc., Korea — Jul 2020 – Aug 2020
Software Engineer Intern

- Developed real-time video comment overlay system for SK hynix's internal video streaming systems
- Won silver prize (2nd) in internship contest

Publications

1. **Jaehyun Ha**, Yongjoo Park, Wook-Shin Han. *CADENZA: Compiling Natural-Language Intent into Task-Specific Operator DAGs for Semantic Query Processing.* SIGMOD 2027.

2. **Jaehyun Ha**, Yongjoo Park, Wook-Shin Han. *CADENZA in Action: Breaking the Monolith with Intent-Dependent Plan Spaces for Semantic Queries.* VLDB 2026 (Demo).

3. Taesung Lee, **Jaehyun Ha**, Byungchul Tak, Wook-Shin Han. *TurboLynx: Schemaless Graph Engine Strikes Back for General-Purpose Analytics.* VLDB 2026.

4. Taesung Lee, **Jaehyun Ha**, Byungchul Tak, Wook-Shin Han. *TurboLynx in Action: A Schemaless Graph Engine for General-Purpose Analytics.* VLDB 2026 (Demo).

5. Wonseok Lee[*], **Jaehyun Ha**[*], Wook-Shin Han, Changgyoo Park, Myunggon Park, Juhyeng Han, Juchang Lee. ([*]equal contribution) *DoppelGanger++: Towards Fast Dependency Graph Generation for Database Replay.* SIGMOD 2024.

6. Wonseok Lee, **Jaehyun Ha**, Wook-Shin Han, Changgyoo Park, Myunggon Park, Juhyeng Han. *DoppelGanger++ in Action: A Database Replay System with Fast Dependency Graph Generation.* VLDB 2024 (Demo).

7. Kyoungmin Kim, **Jaehyun Ha**, George Fletcher, Wook-Shin Han. *Guaranteeing the O(AGM/OUT) runtime for uniform sampling and size estimation over joins.* PODS 2023.

Invited Talks

- Optimizing Semantic Queries: Current Work and Open Questions — Cornell University — Sep 2026
- CADENZA: Intent-Dependent Planning for Semantic Queries — Google Systems Research Group — Jan 2026

Projects

Intent-Driven Semantic Operator Optimization Jun 2025 – Sep 2025

Visiting Scholar Research at UIUC with Prof. Yongjoo Park

- **Problem:** Semantic operators are prohibitively expensive in latency and dollar cost. Existing optimization methods (i.e., proxy models) present a poor trade-off, forcing a choice between unacceptably low accuracy or still-significant latency and cost, which limits practical adoption
- **Contribution:** Designed CADENZA to compile natural-language intent into task-specific plans and optimize their physical implementations
- **Outcome:** Accepted at SIGMOD 2027 and VLDB 2026 (Demo)

High-Performance Schemaless Graph Database System Jan 2023 – Jun 2025

- **Problem:** The conversion of unstructured data into knowledge graphs produces schemaless graphs, where nodes and edges have their own different schemas. However, existing database systems have significant limitations in performing high-performance analytics on such data
- **Contribution:** Designed and implemented a full-stack system with a specialized storage, optimizer, and query engine tailored for schemaless graph analytics
- **Role:** Core developer for a large-scale system (246K+ LOC); authored approx 50% of commits
- **Website:** https://turbolynx.io/
- **Outcome:** Accepted at VLDB 2026 and VLDB 2026 (Demo)

High-Speed Dependency Graph Generation for Database Replay Dec 2022 – Dec 2023

Industry Collaboration with SAP Labs Korea

- **Problem:** Database replay systems are innovative tools for capturing and replaying real-world workloads for testing purposes. However, their real-world efficiency is severely bottlenecked by the slow process of dependency graph generation
- **Contribution:** Proposed an efficient algorithm to accelerate dependency graph generation

- **Impact:** Implemented on the commercial SAP HANA system, demonstrating real-world utility
- **Outcome:** Accepted at SIGMOD 2024 and VLDB 2024 (Demo)

Theoretically Optimal Join Cardinality Estimation Algorithm Mar 2022 – Nov 2022

- **Problem:** Existing sampling-based algorithms for join cardinality estimation lacked a provable optimal O(AGM/OUT) runtime bound
- **Contribution:** Developed the first algorithm to achieve this theoretical optimal bound
- **Outcome:** Accepted at PODS 2023

Technical Skills

Programming Languages	C/C++, Python, SQL, CUDA
Database Systems	PostgreSQL, DuckDB, Kuzu, Neo4j
Systems & Build Tools	Linux, Docker, CMake, Bazel
ML Frameworks	PyTorch, Hugging Face

Honors and Fellowships

- POSTECHIAN Fellowship 2022, 2023, 2024, 2025
- SK hynix Industry-Sponsored Scholar 2020–Present
- Samsung Dream Scholarship Foundation Scholar 2019-2021

Teaching Experiences

- **Teaching Assistant**, Agentic RAG, Samsung Electronics Spring 2026
- **Teaching Assistant**, AIGS540: Big Data Processing, POSTECH Spring 2024
- **Teaching Assistant**, CSED421: Database Systems, POSTECH Spring 2022

References

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POSTECH Distinguished Professor, Pohang University of Science and Technology (POSTECH), Korea

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Assistant Professor, Department of Computer Science, University of Illinois Urbana-Champaign (UIUC), USA